Hazard Map World Cup



Homework- Day One

1. With your team, pick a volcano where you want to make a hazard map.
2. Review the eruption history of your volcano using the Smithsonian Institution GVP website (http://www.volcano.si.edu/) and/or observatory reports.
   - What is typical behavior for an eruption in recent history (VEI, domes, lava flows, lahars, magma type, effusive/explosive)?
   - What are the potential larger, longer-term styles of eruption?
3. Gather information about hazards at your volcano:
   - Lahar, PDC, and ballistics history; volumes; and recurrence frequency (how big? how often?)
   - What are the triggers for lahars (typhoon, rainstorm, earthquake, eruption, landslide, pyroclastic flow, rapid snow melt)?
   - What types of PDCs does the volcano produce? Dome-collapse, column-collapse, lava flow-front collapse, lateral blasts?
   - Where have lahars, PDCs, and ballistics been mapped or reported?
4. Is there already a hazard map?
   - Is there perhaps a good reason to make a new map for a **different scenario, different purpose, or different audience**?
   - Use https://volcanichazardmaps.org/ to look at maps for your volcano and other example maps
5. Discuss what scenario(s) you would like to portray.

Homework- Day Two

1. Review any spatial information about your volcano.
    - Geologic map: where are the pyroclastic flows, lava flows, lahars, etc.
    - Use Google Earth to explore your volcano
    - Where are the steep areas? Are the channels broad, shallow, narrow, deep? What is the summit like? Pointy and conical? Is there a deep crater? Is it open to one side?
    - Make an elevation profile and think about the H/L value you might use for LAHARZ
2. Where are the people?
    - Find a population map. How many people live nearby?
3. Identify major roads, rivers, lakes, villages, major cities. Where are the most important, critical places if a disaster happens?
    - Airports, energy resources, military assets, hospitals, schools and evacuation centers, sensitive natural areas
4. Decide on the scenario or scenarios for hazard map:
    - Are you considering worst-case and/or most-likely scenarios? Or a range?
    - What type of lahars are you portraying (rain-fall triggered, eruption triggered, etc?)
    - What types of PDCs are you portraying (dome-collapse, column-collapse, etc?)
    - What size/volume eruption(s), lahars, and PDCs are you considering?
5. Decide on your target audience:
    - General public? Mining company or transportation agency? Tourists? Emergency managers? Land-use planners?

Things to Consider

- Map should be beautiful and easy to understand.
- Put your names on it. Stake your professional reputation on it.
- Logos for partner agencies; make it authoritative.
- Scale bar, north arrow, location map or spatial context
- Metadata, cite references or make acknowledgements to other data sources
- What is your message?
- Who is your audience? Technical or non-technical?
- Are you making a map focused on hazard level (high-moderate-low, hazards are combined) or focused on the type of hazard (PDC, lahar, etc.)?
- Appropriate colors with good design elements
- Areas of relative hazard easy to understand (high, medium, low) or different hazard processes easy to distinguish
- What is your level of uncertainty?
- Oblique photo and/or 3-D view are optional but helpful
- Be creative and have fun!